

Tracking Daily Inflation in Cambodia

The Plain-Language Guide to How We Measure Daily Prices, Clean Data, and Calculate the Consumer Price Index

What is this book about?

This handbook explains in simple, clear, everyday words how our automated system tracks prices across Cambodia every day. It shows how we collect over 35,500 prices every morning, fix typos, convert US Dollars to Khmer Riel, calculate fair inflation numbers, and predict where prices are heading weeks before official monthly reports come out.

No heavy technical jargon. Just clear explanations, practical everyday examples, and simple math.

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1 Why Cambodia Needs a Daily Price Tracker

1.1 Cambodia Uses Two Currencies at the Same Time

Cambodia is unique because people use two currencies every day: the United States Dollar (USD) and the Cambodian Riel (KHR).

- **Where Dollars are used:** In modern supermarkets (like AEON and Lucky), phone shops, electronics stores, restaurant chains, and online shopping apps, prices are almost always shown in US Dollars.
- **Where Riel is used:** In traditional open markets, street food stalls, local tuk-tuks, and for government utility bills (like your home electricity and clean tap water), prices are quoted in Cambodian Riel.

Because both currencies are used side by side, when the exchange rate moves, prices in shops can change quickly. For example, if a family earns their living in Riel, but buys groceries in a supermarket priced in Dollars, any drop in the value of the Riel means their groceries immediately cost more money.

1.2 What Cambodian Families Spend Most of Their Money On

According to official household surveys from the government's National Institute of Statistics (NIS), Cambodian families spend almost all of their monthly budget on three basic things:

1. **Food and Non-Alcoholic Drinks (44.8%):** Nearly half of every dollar spent goes directly to eating and drinking (rice, pork, fish, cooking oil, and vegetables). For poorer families, food can take more than 55% of their budget!
2. **Housing, Water, Electricity, and Gas (17.1%):** Paying for rent, home power, cooking gas, and water.
3. **Transport and Fuel (12.2%):** Gasoline for motorbikes, diesel for trucks, and bus tickets.

Together, these three categories make up **74% of all household spending**. That means when global fuel prices go up, or when meat and rice prices rise, regular families feel the squeeze immediately.

1.3 Why the Traditional Monthly Survey is Too Slow

In the past, the only way to measure inflation was through official monthly surveys. Government workers would walk into physical markets with clipboards once a month, write down prices by hand, and take them back to the office to calculate.

While this traditional method is thorough, it has three big problems:

1. **It is 3 to 4 weeks late:** The official report for January is usually not published until late February. If food or gasoline prices jump today, leaders at the Central Bank and government ministries do not see the official numbers until a month later.
2. **It only checks prices once a month:** If inspectors visit a store on the 10th day of the month, they completely miss sales, discounts, or price jumps that happen on the 20th or 28th.
3. **It takes a lot of time and people:** Walking to hundreds of stalls in heavy traffic is slow, expensive, and easy to make mistakes when copying numbers.

1.4 Our Daily Solution: Automated Web Intelligence

Our automated system solves this problem. Every single morning at 2:00 AM, while the city is asleep, our software automatically visits 25 online store catalogs, fuel company websites, utility regulators, phone providers, vehicle retailers, and bus portals. It collects over 35,500 real prices every day, cleans them, and calculates today's true inflation rate. Government leaders and researchers can see price changes right now, rather than waiting a whole month.

2 The Data Pipeline Architecture (The Medallion Journey)

2.1 The 3 Refining Steps (The Medallion Approach)

Think of data like metal dug out of the ground. When it first comes out, it is raw and dirty. It needs to be cleaned, melted down, and turned into something pure and useful. We organize our data into 3 simple stages:

The 3 Steps: Raw Data to Final Numbers

Step 1: BRONZE (Raw Store Prices)

- We collect raw text and price tags from 25 data sources and stores every morning at 2:00 AM.
- Exactly as the stores wrote them, typos and all. Nothing is thrown away.
- Stored in our raw database table: `bronze.raw_prices`.

↓ *Clean up text, convert Dollars to Riel, match names, remove errors*

Step 2: SILVER (Clean, Standardized Items)

- Fix spelling and standardize sizes (make sure 1000g = 1kg and 500ml = 0.5L).
- Convert all US Dollar prices into Cambodian Riel using today's official exchange rate.
- Connect Khmer names and English names so the computer knows they are the same product.
- Sort every item into its correct official category (Food, Medicine, Tech, Transport).
- Remove crazy errors (like an item marked down by 99% by mistake).

↓ *Run fair price math, fill in out-of-stock items, calculate inflation*

Step 3: GOLD (Final Daily Inflation Reports)

- Calculate the average price change for each group of items using the fair Jevons rule.
- Fill in missing prices if a product is temporarily out of stock.
- Weight each group by how much real families spend on it (Food gets 44.8%, Transport gets 12.2%).
- Produce the final numbers: National Daily CPI, Core CPI (without food and gas), and a prediction of where the month will finish.

2.2 Detailed Layer-by-Layer Workflows

To ensure the system is a perfect digital twin of the code, we follow these exact operational workflows every single day.

2.2.1 The Bronze Workflow: Raw Ingestion

The goal of the Bronze layer is to capture the reality of the market without any filtering.

1. **Trigger:** Apache Airflow wakes up the `cpi_master_dag` at 2:00 AM.
2. **Scrape:** 25 independent scrapers visit store APIs and web pages using `curl_cffi` to bypass security.
3. **Ingest:** Raw text, raw prices, and timestamps are upserted into `bronze.raw_prices`.
4. **Telemetry:** The system records the number of rows collected per store in `staging.bronze_ingestion_stats` to detect if a store website is down.

2.2.2 The Silver Workflow: The Intelligence Engine

The Silver layer turns messy text into a standardized product catalog.

1. **Text Cleaning:** Strip promotional noise ("Special Offer!") and normalize units (1000ml → 1L).

2. **Currency Standardization:** Convert all USD prices to KHR using the NBC morning exchange rate.

3. **Item Matching:**

- Generate a 768-dimensional AI vector for the product name.
- Search the catalog for the closest match using Cosine Similarity.
- Apply **Spec Guardrails** (Storage, Pack size, Volume) to prevent false merges.

4. **Taxonomy Assignment (AI-First Classification):**

- **Tier 1:** Store Domain Purity (e.g., Telcos → Communication, Petrol → Fuel).
- **Tier 2:** Deterministic Text Rules & Seeds (Regex keyword traps & manual overrides).
- **Tier 3:** Direct AI Engine (Google Gemini Flash via `GeminiKeyPool` multi-key pool, resolving 5-digit UN COICOP 2018 codes with native Khmer/English comprehension and domain guardrails).
- **Tier 4:** Automated Database Memoization (`silver.canonical_items` & `silver.clean_store_prices`).

5. **Final Store:** Standardized records are saved to `silver.clean_store_prices`.

2.2.3 The Gold Workflow: The Econometric Engine

The Gold layer transforms individual price tags into national economic indicators.

1. **Elementary Indexing:** Calculate price ratios against the base period and combine them using the **Jevons Geometric Mean**.
2. **Gap Imputation:** If a product is out of stock, estimate its price using the average movement of its category.
3. **Hierarchical Aggregation:** Combine indices from Item → Subclass → Group → Division → National Headline CPI using Laspeyres weights.
4. **Nowcasting:** Project the remaining days of the month by analyzing 7-day momentum in Food and Transport, FX pass-through drift, and festive shocks.
5. **Splicing:** Link the pipeline's 100-base index to the official National Institute of Statistics (NIS) 2006-base index.
6. **Reporting:** Final results are published to `gold.fct_cpi_daily` and `gold.fct_cpi_nowcast`.

2.3 How Airflow and Cosmos Keep Everything Running on Time

To make sure this entire process happens smoothly without humans needing to press buttons at 2:00 AM, we use an automatic manager called **Apache Airflow** and a tool called **Astronomer Cosmos**:

- **The Master Alarm Clock (`cpi_master_dag`):** Rings every day at 2:00 AM. It starts all 25 store scrapers at the same time. Once they finish and at least 3 sources report good data, it wakes up the next step.
- **The Cleaning Step (`silver_dag`):** Runs our data cleaning models. Thanks to Cosmos, each cleaning task is its own step. If one step has an issue, we can see exactly which one it is, while the rest keep going.
- **The Final Calculation Step (`gold_cpi_dag`):** Calculates today's official daily inflation rate and updates our dashboards before breakfast.
- **The Weekly Housekeeping Step (`cpi_maintenance_dag`):** Runs every Sunday at 1:00 AM to organize database folders for the coming months and keep the system fast.

2.4 Splitting Big Tables into Monthly Folders

In one year, we collect almost 13 million price records! If you dump 13 million rows into one giant table, searching for today's price is like looking for a needle in a massive haystack. It would take several seconds for every click.

Instead, we use **Declarative Range Partitioning**. Think of this as putting each month's receipts into its own labeled physical folder (like `prices_2026_08` and `prices_2026_09`). Our system is proactive: every week, it automatically creates these folders 3 months into the future, so there is never a gap in collection. When someone asks: "*What is today's price?*", the computer ignores all the old folders and only looks inside today's folder. The answer comes back in less than **8 milliseconds**!

3 Deep Dive: Collecting the Prices (Bronze Layer)

3.1 The Stores and Services We Actually Scrape

To calculate an honest and comprehensive Consumer Price Index, our automated system checks over 35,500 prices every morning across 25 real digital data channels in Cambodia.

Deep-Dive: What We Extract From AEON Cambodia

AEON is the largest modern hypermarket and department store operator in Cambodia. Rather than treating it as just a grocery store, our pipeline monitors two distinct flagship portals (`aeon` and `aeon3`), extracting thousands of products across 7 different official consumption categories:

- **Fresh Foods & Meats (Division 01):** Fresh pork chops, beef cuts, chicken, local Tonle Sap fish, prawns, jasmine rice, fresh vegetables (morning glory, Chinese cabbage, tomatoes), fresh fruits (bananas, mangoes, oranges), fresh milk, butter, and chicken eggs.
- **Packaged Foods & Seasonings (Division 01):** Canned fish, instant noodles, cooking oils (palm, soybean), soy sauce, fish sauce, white sugar, salt, black pepper, biscuits, and breakfast cereals.
- **Drinks, Beverages & Alcohol (Divisions 01 & 02):** Clean bottled drinking water, fruit juices, iced teas, soft drinks, local and imported beers (Angkor, Cambodia, Heineken), and cooking/table wines.
- **Household & Kitchen Goods (Division 05):** Laundry detergents, fabric softeners, dishwashing liquids, floor cleaners, kitchen trash bags, frying pans, cooking pots, dining plates, and drinking glasses.
- **Health, Hygiene & Over-The-Counter Medicine (Division 06):** Pain relief tablets (paracetamol), medicated balms, bandages, antiseptic liquids, toothbrushes, and toothpaste.
- **Clothing, Shoes & Apparel (Division 03 via AEON 3):** Men's shirts, trousers, women's dresses, children's clothing, school uniforms, everyday casual shoes, sandals, and socks.
- **Personal Care & Grooming (Division 12):** Bath soaps, shampoos, hair conditioners, body lotions, deodorants, face cleansers, and baby wipes.

3.2 Master Inventory: Stores and Available Items

The table below lists every active data source in our pipeline, its unique system identifier (slug), the official spending categories it covers, and the specific items we collect from it:

Data Source	Store Slug	Official Category	Detailed Items Collected
AEON 1 Supermarket	<code>aeon</code>	Food (01), Drinks (01, 02), Household (05), Health (06), Personal Care (12)	Fresh meats, fish, vegetables, rice, milk, cooking oils, snacks, drinks, beer, dish soap, laundry detergent, pans, pain balms, soaps.
AEON 3 Mean Chey	<code>aeon3</code>	Clothing (03), Footwear (03), Cosmetics (12)	Men's shirts, trousers, women's dresses, children's clothes, shoes, sandals, skin lotions, makeup.
DeliShop Cambodia	<code>delishop</code>	Food (01), Drinks (01, 02)	European gourmet groceries, artisanal bakery, imported French/Italian cheeses, organic meats, olive oils, specialty wines.

Data Source	Store Slug	Official Category	Detailed Items Collected
Lucky Supermarket	grab_lucky	Food (01), Drinks (01), Household (05), Care (12)	Daily fresh produce, dairy, chilled meats, canned goods, Asian sauces, snacks, paper towels, cleaning supplies via GrabMart.
Chip Mong Supermarket	grab_chipmong	Food (01), Drinks (01), Household (05), Care (12)	Fresh beef, pork, poultry, pantry spices, breakfast cereals, cooking oils, soft drinks, dishwashing liquids via GrabMart.
Community Pharmacy	communitypharm	Health (06), Personal Care (12)	Prescription drugs, antibiotics, hypertension/diabetes medicines, cough syrups, antiseptic creams, first-aid bandages.
Ucare Pharmacy	grab_ucare	Health (06), Personal Care (12)	Painkillers, cold/flu medicines, throat lozenges, vitamins, infant milk formulas, baby care products via GrabMart.
Khmer Samnang Phone	samnangshop	Communication (08), Computing (09)	Smartphones (iPhone, Samsung Galaxy, Oppo), iPads, tablets, smartwatches, power banks, fast chargers.
Ary Store Phone Shop	arystore	Communication (08), Audio Tech (09)	Budget and mid-tier smartphones (Xiaomi, Realme, Vivo), wireless bluetooth earbuds, phone accessories.
Smart Axiata Mobile	smart	Communication (08.3.0)	Smart ThomMornng prepaid internet plans, monthly mobile data packages, traveler SIMs, roaming packages.
Smart Home Internet	smart	Communication (08.3.0)	Smart @Home wireless broadband routers, Fiber+ standard and ultra high-speed plans (40–120 Mbps).
Cellcard Mobile	cellcard	Communication (08.3.0)	Prepaid mobile 4G/5G data plans, voice call minutes, tourist SIM cards.
Cellcard Home Internet	cellcard	Communication (08.3.0)	Monthly fixed broadband fiber subscriptions and wireless home gateway plans.
Metfone Cambodia	metfone	Communication (08.3.0)	Metfone 4G/5G mobile prepaid bundles, unlimited data packages, and home fiber internet plans.
Khmer Moto Shop	khmermoto	Motorcycles (07.1.2)	Retail catalog of new and pre-owned motorcycles, motorbikes, scooters (Honda Scoopy, Wave, Click, Dream).
Electricité du Cambodge	edc	Electricity Grid (04.5.1)	Official progressive residential lifeline electricity tariffs across domestic consumption tiers (380 to 730 KHR/kWh).
Phnom Penh Water Supply	ppwsa	Municipal Water (04.4.1)	Official progressive domestic piped tap water tariffs across household consumption tiers (400 to 2,200 KHR/m ³).
MOC / Kampuchea Tela	new_gasoline	Fuels (07.2.2), LPG (04.5.2)	Official retail pump prices for Regular Gasoline, Super 95, Diesel, and domestic LPG cooking gas cylinder refills, extracted primarily via official Telegram announcements and gazettes.

Data Source	Store Slug	Official Category	Detailed Items Collected
Khmer24 Real Estate	khmer24	Housing Rents (04.1.1)	Monthly residential rentals: 1-bedroom apartments, 2-bedroom condos, studio rooms, and shophouses in Phnom Penh.
Realestate.com.kh	realestate	Housing Rents (04.1.1)	Urban residential rents, serviced apartments, and condominium rentals across BKK1, Chamkarmon, and Tuol Kork.
BookMeBus Cambodia	bookmebus	Passenger Transport (07.3.2)	Intercity passenger bus, van, and ferry tickets connecting Phnom Penh, Siem Reap, Sihanoukville, Kampot, and Battambang.
redBus Cambodia	redbus	Passenger Transport (07.3.2)	Interprovincial passenger bus tickets across major operators (Giant Ibis, Laryta Express, Virak Buntham).
Bayon BKK Restaurant	bayonbkk	Dining Out (11.1.1)	Prepared meals, breakfast noodle soups (Kuy Teav), fried rice, traditional Khmer lunch sets, bakery cakes, iced coffee.
Sokha Hotels	sokhahotel	Hotels (11.2.0), Dining (11.1.1)	Hotel accommodation per night (deluxe rooms, suites), restaurant dining, hotel buffet breakfasts.
Hyatt Regency Hotel	hyyathotel	Hotels (11.2.0), Dining (11.1.1)	Luxury hotel room accommodation per night in Phnom Penh, dining services, Sunday brunch.
L192 Marketplace	l192	Clothing (03), Footwear (03), Home (05)	Casual men's and women's apparel, t-shirts, dresses, everyday footwear, fashion bags, lifestyle home goods.
MEF / NBC Exchange	mef_fx	Macroeconomic FX Rate	Official daily USD/KHR market exchange rate used for pipeline-wide dual-currency standardization.
NIS Official CPI	nis_official_cpi	Macro Benchmark	Official monthly benchmark CPI numbers from the National Institute of Statistics used to validate daily nowcasts.

3.3 How We Scrape Each Source (Methods and Technology)

Different websites and mobile applications use completely different technical architectures. To collect clean data reliably without crashes or blocks, our engineering team reverse-engineered the exact native API or protocol used by each source:

Store or Service	Store Slug	Scraping Method & Technology	Data Format & Reliability Strategy
AEON 1 Supermarket	aeon	Next.js Proxy REST API with curl_cffi	JSON payload. Spoofs Chrome 124 TLS cipher handshakes to bypass Cloudflare.
AEON 3 Mean Chey	aeon3	Next.js Proxy REST API with curl_cffi	JSON payload. Traverses category tree to capture exact fashion/beauty items.

Store or Service	Store Slug	Scraping Method & Technology			Data Format & Reliability Strategy
DeliShop Cambodia	delishop	Mobile / Web Catalog REST API			JSON payload. Directly queries store catalog endpoints with pagination.
Lucky Supermarket	grab_lucky	GrabMart	Mobile	Session Gateway	JSON payload. Uses mobile app bearer tokens and consumer location headers.
Chip Mong Supermarket	grab_chipmong	GrabMart	Mobile	Session Gateway	JSON payload. Simulates mobile app requests for live store shelves and discounts.
Community Pharmacy	communitypharmacy	Supabase Database API		PostgreSQL REST	JSON payload. Fetches live product tables directly via PostgreSQL with anonymous public key.
Ucare Pharmacy	grab_ucare	GrabMart	Mobile	Session Gateway	JSON payload. Extracts real-time pharmacy prices and OTC inventory via mobile gateway.
Khmer Samnang Phone	samnangshop	WordPress REST API		WooCommerce	JSON payload. Queries official WooCommerce endpoints (/wp-json/wc/v3/products).
Ary Store Phone Shop	arystore	WordPress REST API		WooCommerce	JSON payload. Traverses paginated WooCommerce product catalog with stock checks.
Smart Axiata	smart	Web Plan Catalog & REST Parser			JSON/HTML. Extracts prepaid mobile, data plans, and fiber tariffs from official plan pages.
Cellcard Cambodia	cellcard	Web Plan Catalog & REST Parser			JSON/HTML. Pulls 4G/5G bundles, monthly fiber packages, and tourist SIM tariffs.
Metfone Cambodia	metfone	Web Catalog Parser with Mobile Emulation			HTML/JSON. Extracts Metfone prepaid data packages and home fiber subscriptions.
Khmer Moto Shop	khmermoto	WordPress REST API		WooCommerce	JSON payload. Pulls retail motorcycle models, specs, and price quotes.
Electricité du Cambodge	edc	Official Tariff Schedule Ingestion			Clean structured gazette tariffs across lifeline domestic electricity tiers.
Phnom Penh Water Supply	ppwsa	Official Tariff Schedule Ingestion			Clean structured gazette tariffs across household municipal water consumption tiers.
Ministry of Commerce	new_gasoline	Tela Gazette Parser	Telegram Mirror		HTML/Text. Extracts official 10-day pump prices for Regular, Super, Diesel, and LPG gas.

Store or Service	Store Slug	Scraping Method & Technology	Data Format & Reliability Strategy
Khmer24 Real Estate	khmer24	HTML DOM Scraping with Rate Limits	HTML parsing. Extracts monthly rental listings with randomized pauses to prevent IP blocks.
Realestate.com.kh	realestate	Next.js JSON API & DOM Traversal	JSON/HTML. Parses urban apartment and condo rental listings with location filters.
BookMeBus Cambodia	bookmebus	BookMeBus Booking REST API	JSON payload. Queries route-specific intercity ticket endpoints across Cambodia.
redBus Cambodia	redbus	redBus Search REST API	JSON payload. Queries interprovincial bus routes and schedules in real time.
Bayon Restaurant BKK	bayonbkk	Apollo GraphQL Client Cache State	Next.js <code>__NEXT_DATA__</code> extraction. Reads hydrated Apollo menu state directly.
Sokha Hotels	sokhahotel	Booking Engine REST API	JSON payload. Queries nightly room rates and dining packages across properties.
Hyatt Regency Hotel	hyyathotel	Global Booking REST API	JSON payload. Extracts room rates per night for standard and club rooms.
L192 Marketplace	l192	Official L192 GraphQL Query API	GraphQL query. Batches product queries with category filters for high-speed extraction.
MEF / NBC Exchange	mef_fx	MEF / NBC Open Data REST API	JSON payload. Fetches official daily market USD/KHR exchange rate at 08:00 AM.
NIS Official CPI	nis_official	Official NIS Benchmark Importer	JSON/CSV. Ingests monthly official benchmark CPI numbers from the National Institute of Statistics used to validate daily nowcasts.

3.4 Visiting Websites Politely Without Getting Blocked

Big e-commerce websites deploy automated defenses (such as Cloudflare or AWS Shield) to protect their servers. Here is how our scrapers operate responsibly and ensure 100% daily collection success:

1. **Impersonating Standard Web Browsers:** Standard Python scripts look robotic to security filters. We use advanced browser emulation (`curl_cffi`) that matches the exact cryptographic TLS handshake (JA3/JA4 fingerprint) of Google Chrome on Windows.
2. **Polite Pauses and Randomized Jitter:** We never hammer a website with dozens of requests at once. Our scrapers include randomized pauses (1 to 2 seconds between pages) so we never add noticeable load to store servers.
3. **Using Mobile and Public API Endpoints:** Whenever possible, we query the lightweight JSON APIs used by mobile apps rather than downloading heavy web pages with advertising and tracking

scripts.

3.5 Never Recording the Same Price Twice (Deduplication)

Sometimes an internet connection drops or a task needs to retry. What happens if our scraper runs twice on the same morning? Does it record double the prices?

No. We enforce a strict unique database rule: each item at each store can only have **one price per calendar day**. If the scraper runs a second time on the same day, it uses an `ON CONFLICT DO UPDATE` command, refreshing today's price in-place rather than creating duplicate records. This guarantees that our daily inflation calculations remain clean, honest, and mathematically sound.

4 Deep Dive: Cleaning and Classifying (Silver Layer)

4.1 The Khmer and English Language Challenge

In Cambodia, retail shelves are multilingual. The exact same bottle of milk might be listed in Khmer script in one store, in English in another store, or mixed together:

- Store A: "ទឹកដោះគោស្រស់ 1L" (Khmer)
- Store B: "Fresh Milk 1000ml" (English)
- Store C: "Meiji Fresh Milk ទឹកដោះគោ 1 Litre" (Mixed)

If a computer just looks at the letters, it thinks these are three completely different products! Our Silver cleaning layer translates common terms, normalizes units (1000ml = 1L, 1000g = 1kg), and cleans the text so the computer understands that they represent the exact same product.

4.2 Text Cleaning and Unit Normalization Rules

Before any vector matching takes place, raw text goes through deterministic cleaning in `pipeline/text_clean.py`:

1. **Strip Promotional Jargon:** Removes marketing noise such as "Special Offer!", "Buy 1 Get 1 Free", "Hot Promo", and "New Arrival".
2. **Harmonize Measurement Units:**
 - Mass: Grams, gm, and g are converted to standard kilograms (kg).
 - Volume: Milliliters (ml), ltr, and liter are converted to standard liters (L).
 - Packaging: Single items, 6-packs, and cases of 24 cans are extracted into explicit item quantities.
3. **Dual Currency Conversion:** Converts all USD prices to Cambodian Riel using the National Bank of Cambodia (NBC) official morning exchange rate:

$$P_{\text{KHR}} = P_{\text{USD}} \times \text{FX}_{\text{NBC}}$$

This price in Riel is permanently saved, ensuring that later calculations never accidentally convert it twice.

4.3 AI Vector Embeddings: Matching Items to Catalog

Every morning, scrapers collect thousands of product listings. To track an item's price over time, we must match today's scraped name against our master catalog of over 40,000 known canonical products (`silver.dim_canonical_items`).

4.3.1 The Mathematical Cosine Similarity

To compare the scraped item vector $\vec{u}$ and a catalog item vector $\vec{v}$, we calculate their **cosine similarity**:

$$\text{Similarity}(\vec{u}, \vec{v}) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|} = \frac{\sum_{i=1}^{768} u_i v_i}{\sqrt{\sum_{i=1}^{768} u_i^2} \sqrt{\sum_{i=1}^{768} v_i^2}} \quad (1)$$

The resulting score ranges from 0.0 (totally unrelated) to 1.0 (identical in meaning).

4.3.2 The 3 Matching Tiers and Decision Rules

In `pipeline/vector_item_matcher.py`, the system applies strict decision boundaries:

1. **Tier 1: High Confidence Match** (Similarity ≥ 0.88) → **APPROVE_MATCH**
2. **Tier 2: Borderline Similarity** ($0.75 \leq \text{Similarity} < 0.88$) → **AI Arbitration** (Gemini Flash judges if the items are the same).
3. **Tier 3: Low Similarity** (Similarity < 0.75) → **SPLIT_NEW** (creates a new product in the catalog).

4.3.3 Crucial Spec Guardrails (Preventing False Merges)

Vector embeddings can be tricked by numbers. *"iPhone 15 128GB"* and *"iPhone 15 512GB"* look identical to AI but have very different prices. To stop this, `vector_item_matcher.py` enforces 4 deterministic guardrails:

1. **Storage Guard:** Rejects matches if electronic memory differs (e.g. 128GB vs 256GB).
2. **Pack Quantity Guard:** Rejects matches if pack sizes differ (e.g. 1 single can vs a 24-can case).
3. **Volume/Mass Guard:** Rejects matches if volume or weight differs by more than 10%.
4. **Flavor Variant Guard:** Rejects matching Diet / Zero Sugar drinks with regular sugary drinks.

4.4 The AI-First Classification Architecture

Every product must be assigned to one of the official UN COICOP categories. To do this accurately, cost-effectively, and reliably, we deploy our **AI-First Classification Ladder**:

1. **Tier 1: Verified Cache Lookup & Seed Overrides:** Direct check of `silver.canonical_items` and manual authority overrides (`silver.coicop_override`). Validated items are returned instantly with zero latency or API cost.
2. **Tier 2: Deterministic Rules & Store Purity Locks:**
 - **Store Purity Locks:** Specialized stores are locked to their domain: Realestate/Khmer24 → Housing (04), PPWSA/EDC → Utilities (04), Bus portals → Transport (07), Pharmacies → Health (06), Telcos → Communication (08).
 - **Critical Traps:** Evaluated with word boundary regular expressions (`\ypattern\y`). For example, hair dryers are mapped to Personal Care (12.1.1) rather than generic kitchen/household appliances (05).
3. **Tier 3: Direct 5-Digit AI Engine (Google Gemini Flash):**
 - **High-Speed Batch Ingestion:** Products are classified in batches of 40 items per API call using Google Gemini Flash via `GeminiKeyPool` multi-key rotation.
 - **Bilingual Khmer & English Context:** Directly processes native Khmer retail strings and transliterated descriptions without separate translation hops.
 - **Direct 5-Digit Taxonomy Mapping:** Resolves complete UN COICOP 2018 5-digit subclasses with structured JSON output and domain guardrails (protecting pet food, alcoholic vs non-alcoholic beverages, medicines vs personal care cosmetics).
4. **Tier 4: Automated Database Memoization:** Every newly resolved canonical classification is permanently written to PostgreSQL, guaranteeing that subsequent appearances of the item resolve with zero cost.

5. **Tier 5: Human Expert Review:** The final safety net for borderline items in `silver.needs_review`, with human corrections feeding back into automated regex rule discovery.

Step-by-Step Example: Gemini AI Classifying Instant Noodles

1. **Input Product:** *"Indomie Mi Goreng Fried Noodles Cup 75g"* (Store: AEON, Category: *"Grocery > Instant Noodles"*, Price: 3,200 KHR)
2. **Resolution Path:** Tiers 1–2 fail. The pipeline invokes the AI engine.
3. **Multi-Attribute Prompt:** Gemini receives the name, store, category, and price.
4. **Structured Classification:** Gemini returns: `{"coicop_code": "01.1.1", "confidence_score": 0.99, "reasoning": "Edible cereal-based food product..."}`
5. **Cache Storage:** The result is saved in `silver.dim_coicop_ai_cache` for instant resolution in future scrapes.

5 The Econometric Engine (Gold Layer)

5.1 The Rules of Fair Price Math

To avoid the **Carli Bias** (where simple averages create fake inflation), we use the **Jevons Geometric Mean**. If an egg goes from \$1 to \$2 (+100%) and back to \$1 (-50%), the simple average says +25% inflation, while Jevons correctly says **0% change**.

5.2 Handling Out-of-Stock Items and Product Upgrades

5.2.1 Out-of-Stock (Class-Mean Imputation)

If a product is missing today, we don't use \$0.00. Instead, we look at the average movement of its category (e.g., if other oils rose 1%, we assume the missing oil rose 1%). If an item is missing for **7 consecutive days**, it is officially retired from the basket.

5.2.2 Product Upgrades (Hedonic Regression Dual-Engine)

When a new electronics model is released at a higher price, we separate the cost of *better features* (RAM, storage, camera) from *pure inflation* using **Log-Linear Hedonic Quality Adjustment** (`pipeline/hedonic_regression`).

- **Primary Solver:** Uses statsmodels OLS when installed.
- **Analytical NumPy Fallback:** Automatically falls back to Ordinary Least Squares via `numpy.linalg.lstsq` with identical formulation, ensuring quality adjustments succeed even in lightweight worker environments.
- **Non-Blocking Execution:** In Airflow, unexpected model skips raise an `AirflowSkipException`, allowing the pipeline to succeed without blocking downstream CPI calculations.

5.3 The 12 Shopping Baskets (UN COICOP)

Cambodia's basket follows the UN standard across 12 divisions, with Food (01) being the largest at **44.8%** of total spending.

Division	Code	What is Inside This Category?	Weight (%)
Food and Non-Alcoholic Drinks	01	Rice, pork, fish, cooking oil, fruit, vegetables, clean bottled water	44.78%
Bread and Cereals	01.1.1	Rice (staple food), noodles, flour, bread	17.23%
Meat	01.1.2	Pork, beef, chicken, duck	8.45%
Fish and Seafood	01.1.3	Fresh fish from Tonle Sap, dried fish, shrimp	7.12%
Milk, Cheese and Eggs	01.1.4	Fresh milk, condensed milk, chicken eggs, duck eggs	1.85%
Oils and Cooking Fats	01.1.5	Palm oil, soybean cooking oil, lard	1.14%
Fresh Fruit	01.1.6	Bananas, mangoes, watermelon, oranges	2.46%
Vegetables	01.1.7	Morning glory, cabbages, tomatoes, garlic, onions	2.38%
Sugar and Sweets	01.1.8	White sugar, palm sugar, honey	0.72%
Sauces and Seasonings	01.1.9	Fish sauce, soy sauce, salt, MSG, black pepper	0.63%
Drinks (Non-Alcoholic)	01.2	Bottled water, iced coffee, tea, fruit juices	2.80%
Alcohol and Tobacco	02	Beer (Angkor, Cambodia), wine, spirits, cigarettes	1.63%

Division	Code	What is Inside This Category?	Weight (%)
Clothing and Footwear	03	Men's and women's shirts, trousers, school uniforms, shoes	3.04%
Housing, Water, Power, Gas	04	Home rent, clean city water (PPWSA), electricity (EDC), cooking gas	17.08%
Tenant Rent	04.1.1	Actual rent paid for rooms and apartments	9.42%
City Water	04.4.1	Municipal piped tap water bills	1.86%
Electricity	04.5.1	Home power bills from EDC	2.82%
Cooking Gas	04.5.2	LPG 15kg cooking gas cylinder refills	1.65%
Furniture and Household Goods	05	Beds, chairs, kitchenware, laundry detergent, cleaning soap	3.25%
Health and Medicine	06	Essential medicines, pain relievers, clinic visits, doctor checks	5.56%
Medicines and Pharmacy	06.1.1	Paracetamol, antibiotics, cough medicine	3.45%
Medical Consultations	06.2.1	Basic clinic visits and health checkups	1.74%
Transport and Gasoline	07	Motorbikes, gasoline (Regular, Super, Diesel), city buses	12.18%
Motorcycles	07.1.2	Popular motorbikes (Honda Dream, Wave, Scoopy)	3.12%
Gasoline and Fuels	07.2.2	Daily pump prices at PTT, Tela, Total	6.85%
Bus and Van Travel	07.3.2	Passenger bus tickets between provinces	1.65%
Communication	08	Mobile phones, SIM cards, monthly internet packages	3.92%
Smartphones	08.2.0	Popular handsets (iPhone, Samsung, Vivo, Oppo)	1.42%
Mobile Data Plans	08.3.0	Smart, Cellcard monthly 4G/5G data plans	2.50%
Recreation and Culture	09	Televisions, laptops, stationery, school books	1.91%
Education	10	School tuition fees, English classes, university credits	1.51%
Restaurants and Eating Out	11	Street food breakfast, local cafes, restaurant meals	3.09%
Personal Care and Other Goods	12	Haircuts, soap, shampoo, cosmetics, basic personal care	2.07%
Total National Basket	ALL	Complete Cambodian Household Spending	100.00%

5.4 Hierarchical Aggregation: From Item to National CPI

A national CPI is a pyramid. We combine data from the bottom up:

1. **Level 1 (Items):** Collect daily prices for thousands of products.
2. **Level 2 (Subclasses):** Combine items into groups like "*Bread and Cereals*" using the Jevons Mean.
3. **Level 3 (Groups):** Combine subclasses into larger groups using expenditure weights.
4. **Level 4 (Divisions):** Combine groups into the 12 Major Divisions.
5. **Level 5 (National Headline CPI):** The final weighted sum of all 12 divisions.

5.5 Predicting the Rest of the Month (Nowcasting)

Official reports are 4 weeks late. Our `ml/nowcaster.py` engine predicts the monthly outcome in real-time using 10 mathematical equations.

- **Momentum:** It checks the 7-day price velocity of Food and Transport.
- **FX Drift:** It calculates how much the USD/KHR exchange rate is pushing prices up.
- **Festive Shocks:** It adds known demand surges for Khmer New Year and Pchum Ben.
- **Blended Result:** It combines observed days with the projected remaining days to give a **Nowcast CPI** with a 95% Confidence Interval.

6 The Mathematical Foundation (The 24 Formulas)

Every mathematical equation used in the pipeline has a clear purpose. Here is each formula explained in plain English.

6.1 Core Indexing Formulas

- **Eq 1: Base Price ($P_{i,0}$):** The geometric average price over the base period.
- **Eq 2: Price Ratio ($R_{i,t}$):** Today's price divided by the base price.
- **Eq 3: Outlier Bound:** Blocks prices that jump more than $5\times$ (prevents typos from ruining the index).
- **Eq 5: Imputation ($\hat{P}_{i,t}$):** Estimates missing prices using group momentum.
- **Eq 6: Jevons Index ($I_{J,c}$):** The fair average price change for a group of items.
- **Eq 8: National Headline CPI:** The final weighted sum of all 12 divisions.
- **Eq 9: Core CPI:** The inflation rate excluding Food, Utilities, and Transport.

6.2 AI and Quality Formulas

- **Eq 11: Cosine Similarity:** Measures the semantic meaning between two product titles.
- **Eq 12: Token-Sort Fuzzy Similarity:** Handles different word orders (e.g., "*Beer Angkor*" vs "*Angkor Beer*").
- **Eq 14: Hedonic Regression:** Measures how much extra shoppers pay for better hardware (RAM, Storage).
- **Eq 15: Constant-Quality Price:** Subtracts the value of tech upgrades to isolate pure inflation.

6.3 Nowcasting Formulas

- **Eq 16: Leading Drift ($\hat{\delta}_{\text{leading}}$):** Combines trailing 7-day momentum from Food and Transport.
- **Eq 17: FX Drift ($\hat{\delta}_{\text{FX}}$):** Measures inflation caused by the USD/KHR exchange rate.
- **Eq 18: Festive Shock (ϕ_{fest}):** Adds temporary price spikes during national holidays.
- **Eq 21: Blended Nowcast:** Combines observed days with projected remaining days.
- **Eq 24: NIS Splicing:** Connects the pipeline result to the official National Institute of Statistics (NIS) 2006-base index.

7 Macroeconomic Integration: Currency and History

7.1 The Dual-Currency Impact

Because Cambodia uses both currencies, the exchange rate directly changes prices.

- **Supermarkets (USD):** Prices rise almost immediately (within 24 hours) when the Riel drops.
- **Local Markets (KHR):** Prices adjust slowly (about 2 weeks) as vendors absorb the change.
- **Utilities (KHR):** Rates are fixed by the government and do not change with the exchange rate.

7.2 Connecting to Official National History (Splicing)

Our pipeline starts from 100.0, but the government's official NIS series starts from 2006. To make our numbers useful, we use **Splicing**:

$$\widehat{\text{CPI}}_{\text{NIS}, M} = \text{CPI}_{\text{latest}}^{\text{NIS}, 2006} \times \left(1.0 + \frac{\pi_{\text{MoM}}}{100.0}\right) \quad (2)$$

This ensures that a $\text{CPI} \approx 105.8$ in our system translates to $\text{CPI} \approx 220.5$ on the government scale.

8 System Reliability: Operational Excellence and Scale

8.1 The Self-Learning Feedback Loop

Our system learns from human expertise. When an analyst corrects a product category, the `automated_rule_suggester` finds the pattern and proposes a **Regex Rule**. Once approved, this rule is added to the database, so the system never makes the same mistake twice.

8.2 Automated Quality Assurance

To ensure zero errors, we execute:

- **56 dbt Data Tests:** Check that weights sum to 100%, tariffs match gazettes, and no price is negative.
- **451 Python Tests:** Validate API parsing, TLS spoofing, and formula precision.

If any critical test fails, the Airflow pipeline halts immediately, ensuring that corrupted data never reaches the official reports.

8.3 Database Performance at Scale

With 13 million records a year, we use **Declarative Monthly Partitioning**. Data is stored in monthly folders (e.g., `raw_prices_2026_09`). This allows PostgreSQL to use *partition pruning*, skipping 95% of historical data and returning query results in under **8 milliseconds**.

9 Data Consumption: Analytics and Government Use

9.1 Real-Time Executive Dashboards in Metabase

We provide three enterprise dashboards for full transparency:

1. **Macro Analytics Dashboard:** Tracks Headline and Core CPI and compares nowcasts against official NIS benchmarks.
2. **Pipeline Telemetry Dashboard:** Monitors the 25 scraper sources and the daily ingestion matrix.
3. **Silver Data Quality Screener:** Audits COICOP distribution and verifies that prices adhere to ILO outlier boundaries.

9.2 How Government Leaders Can Use This Data

- **Central Bank:** See inflation *now* rather than waiting 4 weeks. Distinguish between global shocks (gasoline) and domestic pressure (food) to apply the correct monetary policy.
- **Ministry of Finance:** Protect vulnerable families by adjusting cash support programs the moment food prices spike.
- **Utility Regulators:** Monitor that water and electricity tariffs remain affordable for everyday citizens across all provinces.

9.3 Conclusion

The **Cambodia Daily Consumer Price Index System** shows that we can track inflation across an entire country every single day. By using automated web scraping, smart and fair math, sensible guardrails, and clear data cleaning, we can give policymakers, researchers, and families up-to-the-minute inflation numbers without spending millions of dollars on manual surveys.

— End of Plain-Language Handbook —